

SHIVAM KUMAR

shivamkm07@gmail.com | +91-9129085031 | [LinkedIn](#) | [GitHub](#) | Hyderabad, India

PROFESSIONAL SUMMARY

SDE II at Microsoft building Azure cloud-native platforms, distributed systems, and Kubernetes-based container infrastructure. Expertise in microservices architecture, AI/LLM-powered SRE automation, and Dapr (CNCF Graduated). IIT Kanpur, 5 YoE.

WORK EXPERIENCE

Software Development Engineer II — Azure Container Apps

Jul 2024 – Present

Microsoft

Hyderabad, India

- Engineered AI-powered SRE agent for Azure Container Apps, reducing P50 TTM by 97% and TTR by 64%; added RCA subagents, self-repair, auto-resolution, and Ev2 deployment pipelines.
- Enhanced Azure Functions on ACA with graceful drain, key management, KEDA scale rules, tracing, and durable event logs; deployed to ~1,322 apps.
- Resolved P0 blockers for ACA Express Public Preview on Azure Dev Compute (ADC), unblocking ~100K+ environments for ephemeral volumes and Log Analytics customer logs.
- Implemented ACA Sessions management APIs, fixed async execution bugs for ~300 external customers, and enabled early Fabric session deletion to reduce resource cost.
- Led Dapr AKS Extension rollouts across ~400 AKS clusters and integrated ACA agentic apps with Foundry Control Plane for central management of ~2,000+ apps.

Software Development Engineer I — Dapr

Jul 2021 – Jul 2024

Microsoft

Hyderabad, India

- Stabilized Dapr building blocks across Configuration API, MQTT PubSub, CosmosDB state, and Redis/PostgreSQL config stores; added Event Hubs bulk-subscribe and CLI airgap install.
- Implemented Cascade Terminate/Purge for Dapr Workflows and fixed scalability bottlenecks found through performance tests.
- Co-launched ACA Sessions, owning Node.js/.NET code-interpreter support for Security Copilot; fixed critical bugs and added regression coverage for 3000+ active sessionPools.
- Designed patch-based internal fork infrastructure across 5 Dapr repositories; built OneBranch/Ev2 pipelines and automated OSS-to-internal releases with <1 day SLA.
- Delivered multi-architecture Dapr/dashboard images, 1P Container Registry migration, ACA nightly builds, and Mariner/distroless images to MCR.

Quantitative Research Intern — E-Trading

May 2020 – Jul 2020

JP Morgan & Chase

Mumbai, India (Remote)

- Built price prediction models using LSTM, feedforward neural networks, and regression with univariate feature selection; improved predictive power by 40% for HK names through feature standardization.

RESEARCH

Data Race Detection for Task-Parallel Programs

Jun 2019 – Aug 2021

Supervisor: Prof. Swarnendu Biswas, IIT Kanpur

- Designed Tasker, combining vector clocks with tree-based techniques; built an LLVM memory instrumentation pass and achieved 1.48x speedup over PTRacer with lower memory overhead.
- “Efficient Data Race Detection of Structured Task Parallel Programs Using Vector Clocks” — S. Kumar, A. Agrawal, S. Biswas | Submitted to ISCGO 2022.

PROJECTS

GemOS | C, x86 — OS kernel (IIT Kanpur CS330) | [GitHub](#)

- Implemented mmap/munmap/mprotect with lazy allocation and page-fault handling; cfork/vfork with copy-on-write semantics.

Cipher Decoder | Python, Modern Cryptology (CS641) | [GitHub](#)

- Performed differential cryptanalysis of 3-DES and decrypted Caesar/Vigenere ciphers using frequency analysis.

TECHNICAL SKILLS

Languages: Go, C#/.NET, Python, C/C++

Cloud & Infra: Azure (ACA, Functions, AKS), Kubernetes, Docker, Helm, KEDA

Distributed Systems: Dapr (CNCF), Microservices, Service Mesh, gRPC, Pub/Sub

EDUCATION

Indian Institute of Technology, Kanpur

2017 – 2021

Bachelor of Technology in Computer Science and Engineering | CGPA: 9.4/10.0

ACHIEVEMENTS & RECOGNITION

- IDC Pinnacle FY'26 – Best AI Success Story (ACA RCA Agent squad, ~200+ competing teams); RockStar Award, Microsoft ('22, '23).
- Academic Excellence Award (x2), IIT Kanpur | JEE Adv. AIR 715 | JEE Main AIR 348 | KVPY Scholar | NSE Physics Top 1%.